Nathan Sauldubois

06 25 96 59 80 — nathan.sauldubois@free.fr — nathansauldubois.github.io

Summary —

I am a postdoctoral researcher at **New York University (NYU)**, working on probability, optimal transport, martingale optimal transport, and model risk. I obtained my **PhD in Applied Mathematics from École Polytechnique** under the supervision of **Nizar Touzi**, and defended my dissertation on **November 25, 2025**. My research combines theoretical probability, optimal control, optimal transport, and distributionally robust optimization.

Skills

Mathematical Interests: Optimal transport, martingale optimal transport, probability, optimal control, mean-field games, robustness

Numerical Interests: Machine learning, deep learning for PDEs, entropic regularization, particle methods
Programming: Python, R, OCaml, LaTeX
Languages: French (native), English (fluent)

Publications

Published Papers & Preprints

- Sauldubois, N., and Touzi, N. (2024). *First-Order Martingale Model Risk and Semi-Static Hedging*. arXiv preprint arXiv:2410.06906. Under revision at *SIAM Journal on Financial Mathematics*.
- Compoint, A., Sauldubois, N., and Touzi, N. (2025). *Sensitivity Analysis of Distributionally Robust BSDEs and RBSDEs*. arXiv preprint arXiv:2511.01828. Under revision at *Mathematical Finance*.
- Sauldubois, N. (2026). *Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach*. arXiv preprint arXiv:2601.20139.
- Sauldubois, N., and Zhang, X. (2026). *Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport*. arXiv preprint arXiv:2605.30560.

In Preparation

- Sauldubois, N. (2024). *Order-n Expansions in Classical and Causal Distributionally Robust Optimization*.
- Sauldubois, N., & Touzi, N. (2023). *A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem*.
- De March, H., Sauldubois, N., & Touzi, N. (2022). *Optimal Transport Meets Healthcare Optimization*.
- Crowell, R., Sauldubois, N., & Touzi, N. (2025). *Distributionally Robust Optimization in Continuous Time Under Law Constraints* .

Talks

- "A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem", Séminaire Doctorant du CMAP (2023)
- "Extended L-Q Mean-Field Game with Common Noise", Hammamet "Stochastic Control and Risk" (April 2023)
- "Robust Hedging and Distributionally Robust Sensitivity", PhD Talk, Séminaire Bachelier (March 2024)
- "Model Risk and Semi-Static Hedging", Bachelier Finance Society Meeting (July 2024)
- "Model Risk and Semi-Static Hedging", Young Researchers Workshop, Berlin (2024)
- "Robust Static Hedging and Distributionally Robust Sensitivity", Hammamet Workshop, "Stochastic Control and Games for Risk and Regulation" (October 2024)
- "Adapted Wasserstein Gradient Flow and Projected Langevin Dynamics for Weak Optimal Transport", Hammamet "Stochastic Control and Risk" (April 2026)
- "Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach", XIII Bachelier World Congress, University of Bologna (July 1, 2026)
- "Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport", DMV Annual Meeting 2026, University of Konstanz (September 9, 2026)
- "Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport", 10th Eastern Conference on Mathematical Finance, Johns Hopkins University, Baltimore (October 2026)

Teaching

Courses, Exercise Sessions, and Tutorials

- **Option Pricing & Stochastic Calculus** — Graduate course, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Spring and Fall 2026)
- **Financial Computing Recitations** — Graduate recitations, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Fall 2026)
- **Summer Bootcamp** — Graduate bootcamp, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Summer 2026)
- **Winter Math Bootcamp: From Brain Teasers to Black–Scholes** — Graduate bootcamp, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Winter 2026)

- **Probability Refreshers** – Master X–HEC Data Science for Business (2022–2025)
- **Stochastic Models in Finance** – École Polytechnique (2022–2025)
- **Convex Optimization and Control** – Third-year bachelor's course (2022–2025)
- **Preparatory Classes (CPGE)** – Oral exam preparation tutor at Lycée du Parc (2019–2020)

Supervision

- Co-supervision of the research internship of **Arthur Compoint** on *continuous-time model risk for stochastic optimization problems* (2025).

Administrative Responsibilities

- Responsible for communications and website maintenance for the **Séminaire Bachelier** (CMAP–CREST)
- Co-organizer of the **Séminaire Doctorant du CMAP**, 2023–2024
- Co-organizer with Xin Zhang of the Invited Session "Optimal Transport in Robust Finance" at the 2026 INFORMS Annual Meeting

Education

École Polytechnique 2022–2025
PhD in Applied Mathematics
Defended on November 25, 2025. Supervisor: Nizar Touzi.

ENS de Lyon 2020–2021
M2 Advanced Mathematics

ENS de Lyon 2019–2020
M1 Advanced Mathematics

ENS de Lyon 2018–2019
L3 Fundamental Mathematics with a Physics Minor

Research Experience

Postdoctoral Researcher, NYU 2025–Present

- Distributionally robust optimization in continuous time.
- Adapted Wasserstein gradient flows and applications to entropic martingale optimal transport and weak optimal transport.

CMAP, École Polytechnique 2022–2025
PhD Research

- Mean-field games with control-dependent interactions.
- Sensitivity analysis and distributional robustness in stochastic control.
- Martingale optimal transport and model risk.

Natixis – Quantitative Research 2021–2022
Research Internship supervised by Pierre Henry-Labordère

- Model uncertainty in fixed-income products.
- Super-replication strategies for Bermudan swaptions with non-constant notionals.
- Quantum optimal transport methods for correlation swaps.

CMAP, École Polytechnique 2021
Research Internship supervised by Nizar Touzi and Hadrien De March

- Entropic optimal transport applied to healthcare policy design, in partnership with Quantev.

Oxford University 2020
Research Internship supervised by Julien Berestycki

- Analysis of high-dimensional branching Brownian motion and extremal particle estimates.

Université de Bordeaux 2019
Research Internship supervised by David Lannes

- Study of shallow-water wave equations and the behavior of solutions in bounded-depth domains.

Additional Activities

- Special Jury Prize – SMF Junior Contest (2018)